

Space-Time Geostatistics for El-Salvadorian Murder Rates

Preprocessing

The data were supplied in an unspecified geographic projection. For the purposes of presentation, the data were matched to a set of polygons containing information on El Salvadorian administrative units. The vector map of El Salvador was known to be in the Universal Transverse Mercator projection (zone 16 North) with the WGS1984 datum. I generated a regular grid sample from these polygons to produce the simulation grid (Fig 1) used for subsequent analysis.

The murder rate data were translated horizontally and vertically to minimise the mean distance measured between the observation coordinates and the administrative unit polygon centroids, using a simple optimisation routine. The resulting translation involved subtracting 215974 from the East coordinates and adding 1228266 to the North coordinates. Obviously a proper affine transform would have been preferable to this simple translation, but without the original projection information for the data the results would have been rather subjective. As such, the matching between observations and administrative units is likely to be imperfect, and any reference to province names in this document is given with caution.

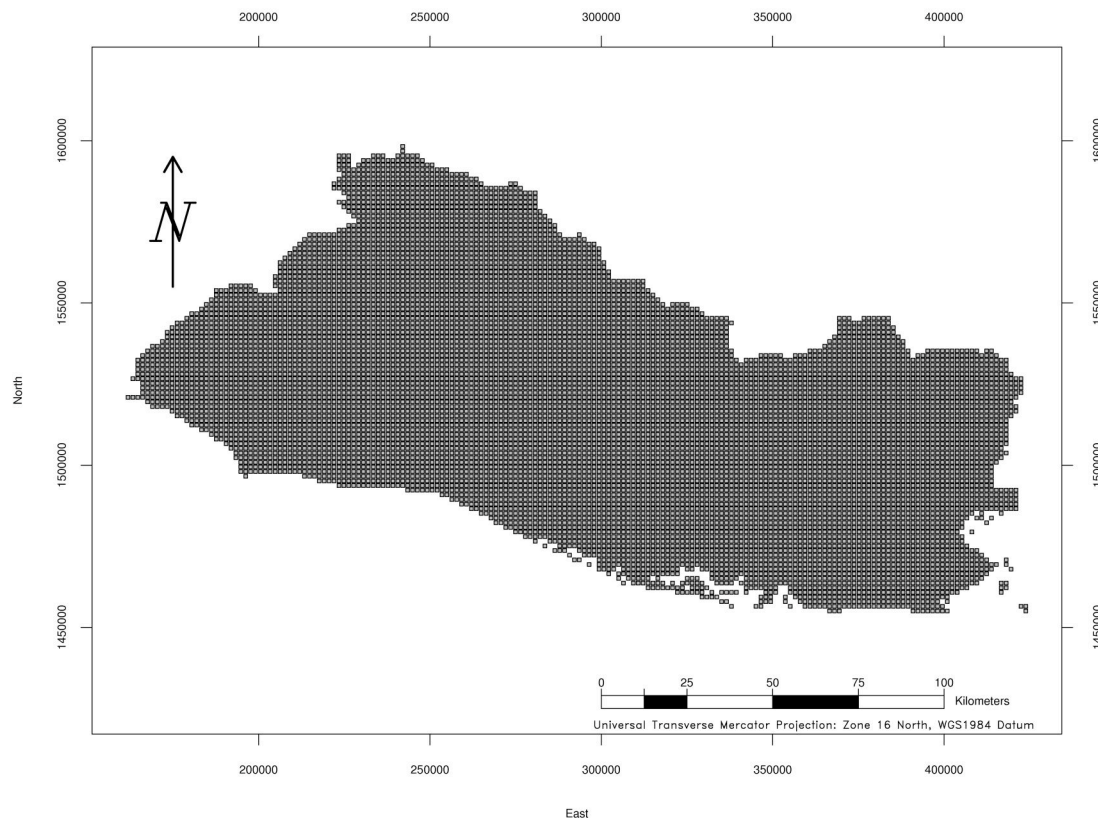


Fig 1: Simulation grid of El Salvador

The only additional processing was to add a very small random deviate to the coordinates to prevent matrix singularities when simulating, which would have arisen if a data point coincided with a grid point.

Data Distribution

The data are strongly left skewed, with some provinces having particularly high rates; most provinces, however, have very low rates, with a mean murder rate of 13.96 for the whole country. However, given the

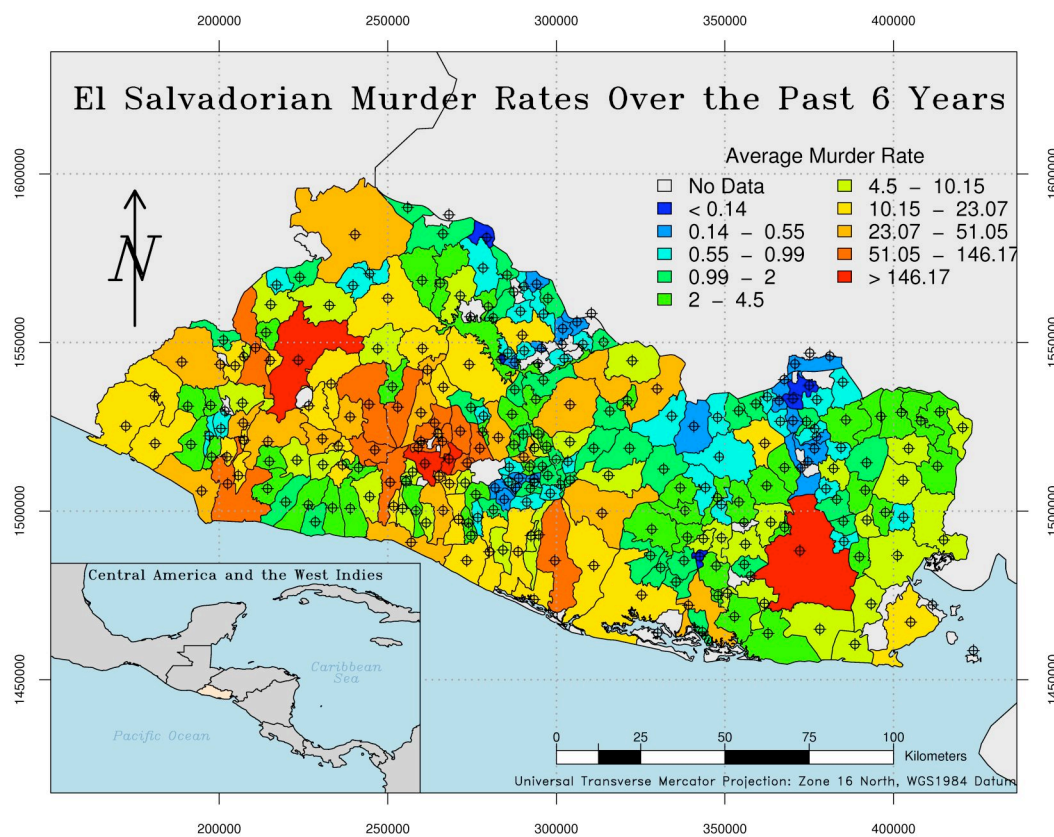


Fig 2: Mean murder rates for El Salvadorian provinces from six years of data

skew in the data set, the median is probably a better indication of central tendency, indicating a background murder rate of 3 murders year⁻¹ for the entire region.

The highest murder rates were observed in San Salvador (SID 111), where 431 cases were recorded for years 1 and 3. There were seven SIDs with murder rates >100 year⁻¹; Sonsonate, Colón, Apopa, San Miguel, Soyapango, Santa Ana, and San Salvador (SID 40, 77, 98, 215, 114, 22 and 111). A map of mean murder rates by province is presented in Fig 2.

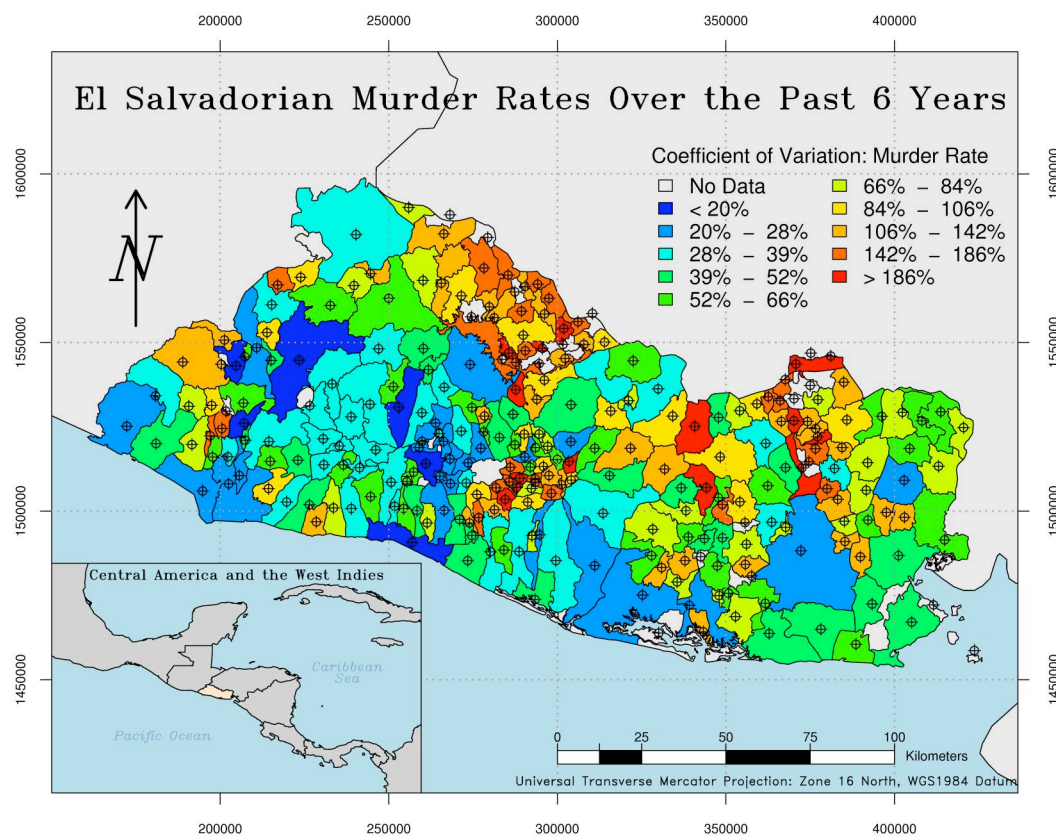


Fig3: Coefficient of variation in murder rates for El Salvadorian provinces from six years of data

An interesting feature of the data is that the provinces with high murder rates seem to have low variances and vice – versa. This would indicate relatively stable and high rates in some ‘problem’ provinces, whilst most provinces have low rates with occasional atypical peaks in murder rate. The heteroscedasticity is evident when looking at the coefficient of variation by province (Fig 3). Fig 4 highlights the atypical provinces by illustrating the data distribution for each province by north-south and east-west transects. It is clear from the data plot that problem provinces are

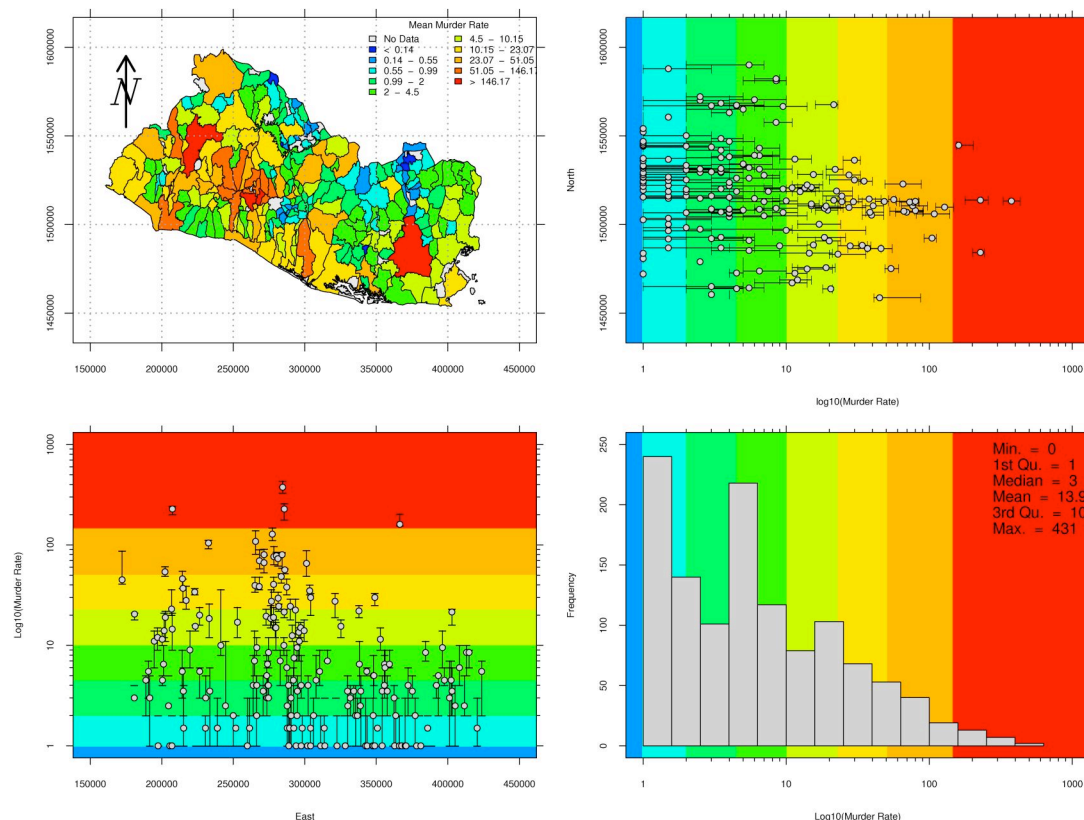


Fig 4: Spatial distribution of high murder rates and their variances. High murder rates are associated with low variances and vice - versa

centred around the capital, with an additional problem province in San Miguel to the East.

Spatial Clustering in Murder Rates

Further examination of the 6-year averages of provinces indicates significant spatial clustering of murder rates. Examining the log transformed data for spatial clustering via Moran's I statistic indicates positive autocorrelation: The log transform is necessary as Moran's I assumes the mean is a good model of the background murder rate, an assumption which

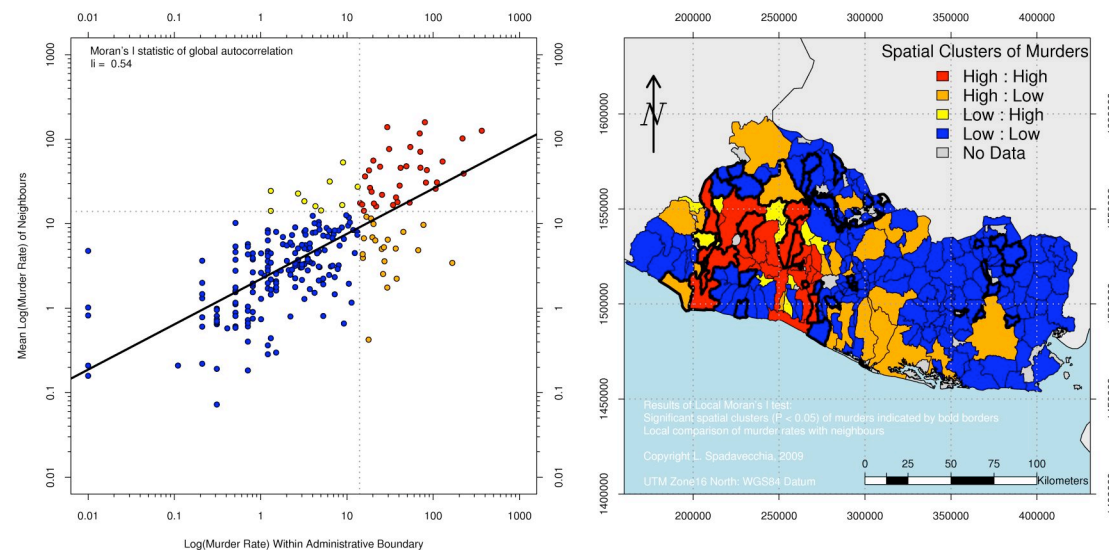


Fig 5: Local indicator of spatial autocorrelation. Significant clusters are bordered in bold lines, whilst colours indicate the type of clustering observed

is violated by the left skew of the raw data. A local indicator of spatial autocorrelation is readily calculated for Moran's I , by comparing the observed murder rate for each province by the mean rate of its immediate neighbours. The apparent clustering around the capital is confirmed by this approach (Fig 5).

Semivariograms

In order to simulate murder rates for next year it is necessary to model the spatio-temporal covariance structure of the data in order to enforce the observed patterns of variation on the predicted surfaces. The product-sum covariance model used to summarise the spatio-temporal autocorrelation structure of the data is derived from a spatial semivariogram, a temporal semivariogram and the global sill variance.

Table 1: Product-Sum covariance model used to draw simulations

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*** Covariance model is permissible. ***
=====

  Spatial Model:
-----
Exponential:      Contribution = 1122.7   Range = 3968.04
Hole Effect:      Contribution = 233.96   Range = 42835.55

  Temporal Model:
-----
Nugget Effect:    Contribution = 19.89
Exponential:      Contribution = 223.65   Range = 22.98

  Sills:
-----
Spatial Partial Sill = 1356.68
Temporal Partial Sill = 243.54
Global Sill          = 1380.92

  Scaling constants (k):
-----
k[1] = 0
k[2] = 0.84
k[3] = 0.1
=====

```

The semivariogram in space is calculated from the variance of all data points separated by given lag distances in space ($\Delta h_{\text{space}} = \{0, 10, 20\text{km } \dots\}$), with a zero lag separation in time ($\Delta h_{\text{time}} = 0$).

The semivariogram in time is calculated from the variance of all data points separated by given lag distances in time ($\Delta h_{\text{time}} = \{0, 1, 2\text{years}\dots\}$), with a zero lag separation in space ($\Delta h_{\text{space}} = 0$).

The global sill is the variance the data would exhibit if sub-sampled such that autocorrelation was zero, i.e. the variance of all pairs of data with

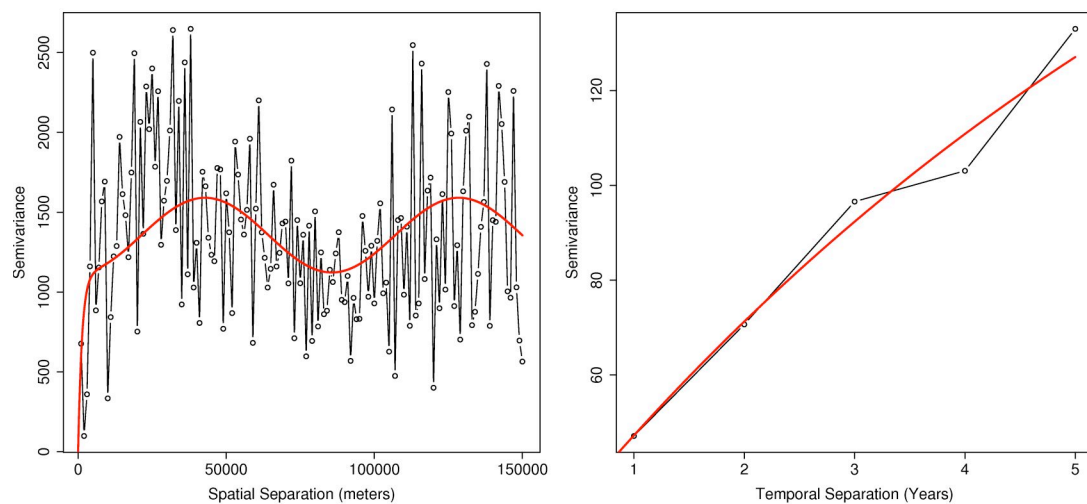


Fig 6: Semivariograms in space and time for the El Salvadorian murder rate data

a Δh_{space} and Δh_{time} greater than the maximum range of autocorrelation in space and time.

The fitted semivariogram model is detailed above (Table 1). The hole effect in the spatial semivariogram indicates that there is a spatial periodicity in murder rates, which is likely due to the leverage of the two spatial clusters of high murder rates around San Salvador and San Miguel. The wavelength of the hole effect (Fig 6) seems to be about 100 – 120km, which is consistent with the separation of the two high clusters.

Simulations

Having defined a permissible covariance model to draw simulations with, it is possible to generate possible scenarios for the time period of the observations. The simulations seem to adequately capture the spatial and

temporal patterns of the data (Figs 7 - 11), indicating that the covariance model is well specified. Drawing simulations for next year is also possible, with 5 realisations illustrated in Fig 12. If a large enough number of draws are made ($n \sim 1000$), the mean and variance of each province can be calculated. The mean of the simulations will be equivalent to the kriging estimate, but the sample variance of the simulations will be a more valid expression of the uncertainty in estimates than the kriging variance, since the data display significant heteroscedasticity. An example is presented in Fig 13, although this is based on a small number of simulations ($n = 100$), and should not be taken as anything other than an illustration.

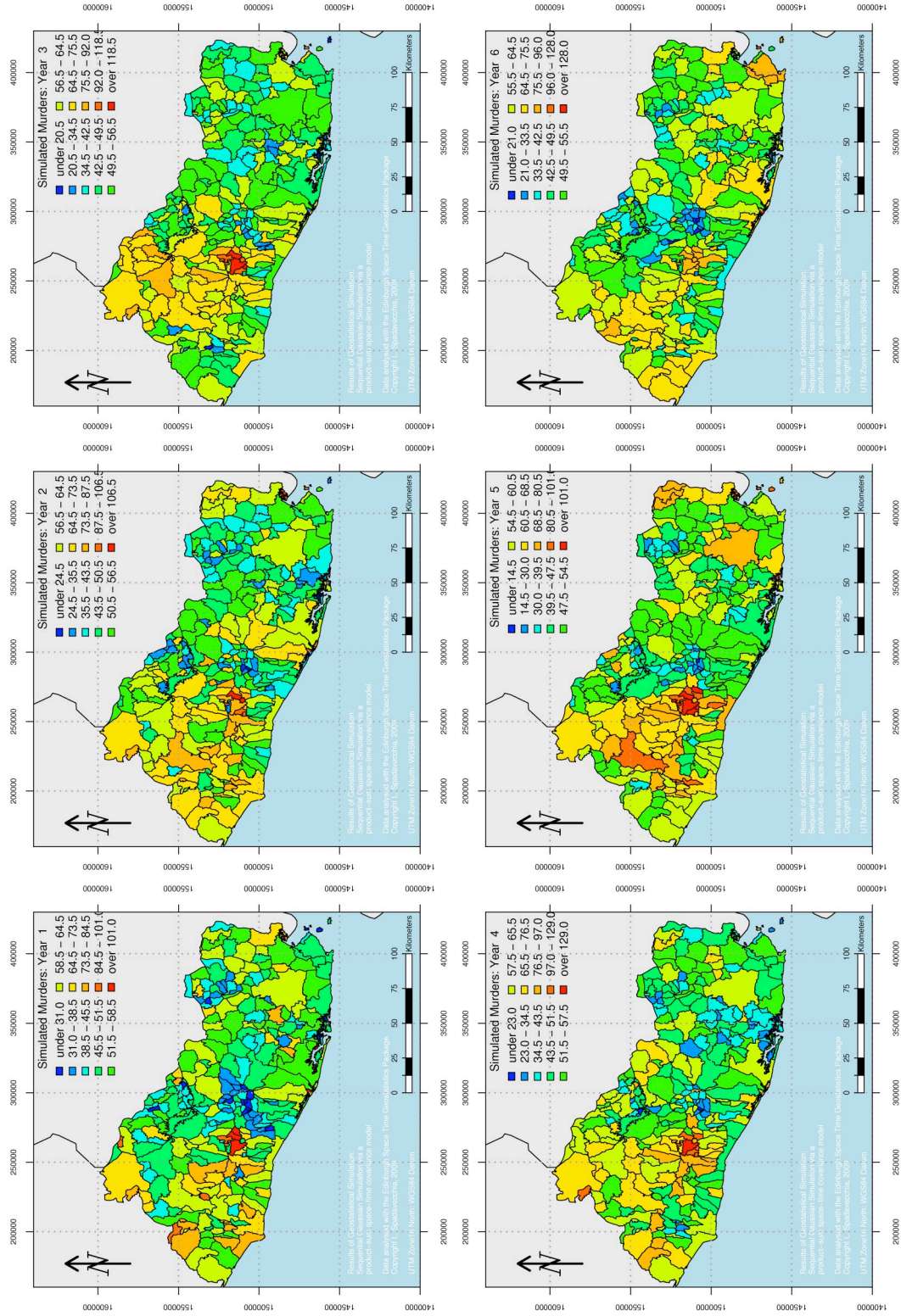


Fig 7: Simulated murder rates for El Salvadorian provinces over 6 years (Simulation 1)

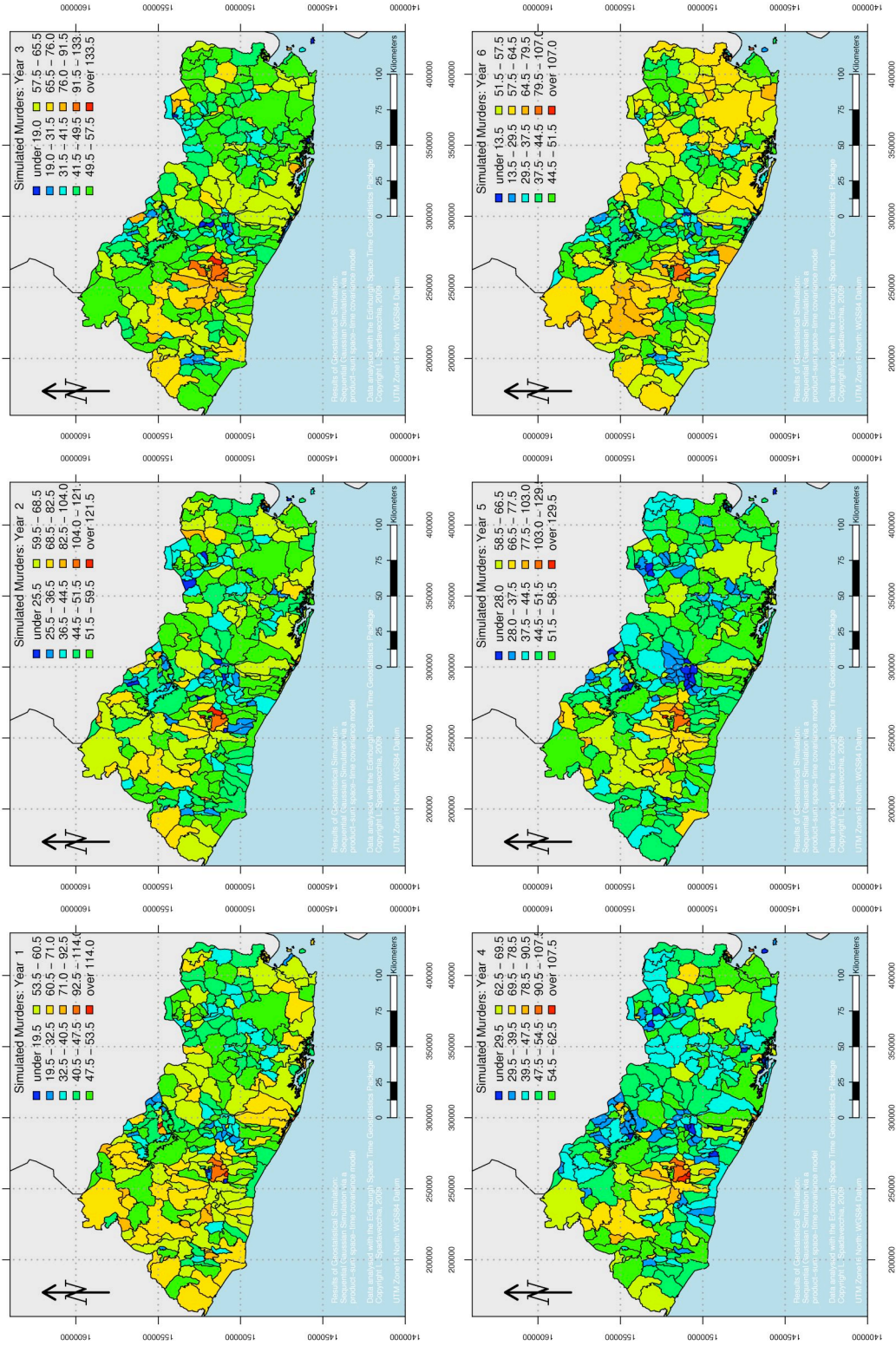


Fig 8: Simulated murder rates for El Salvadorian provinces over 6 years (Simulation 2)

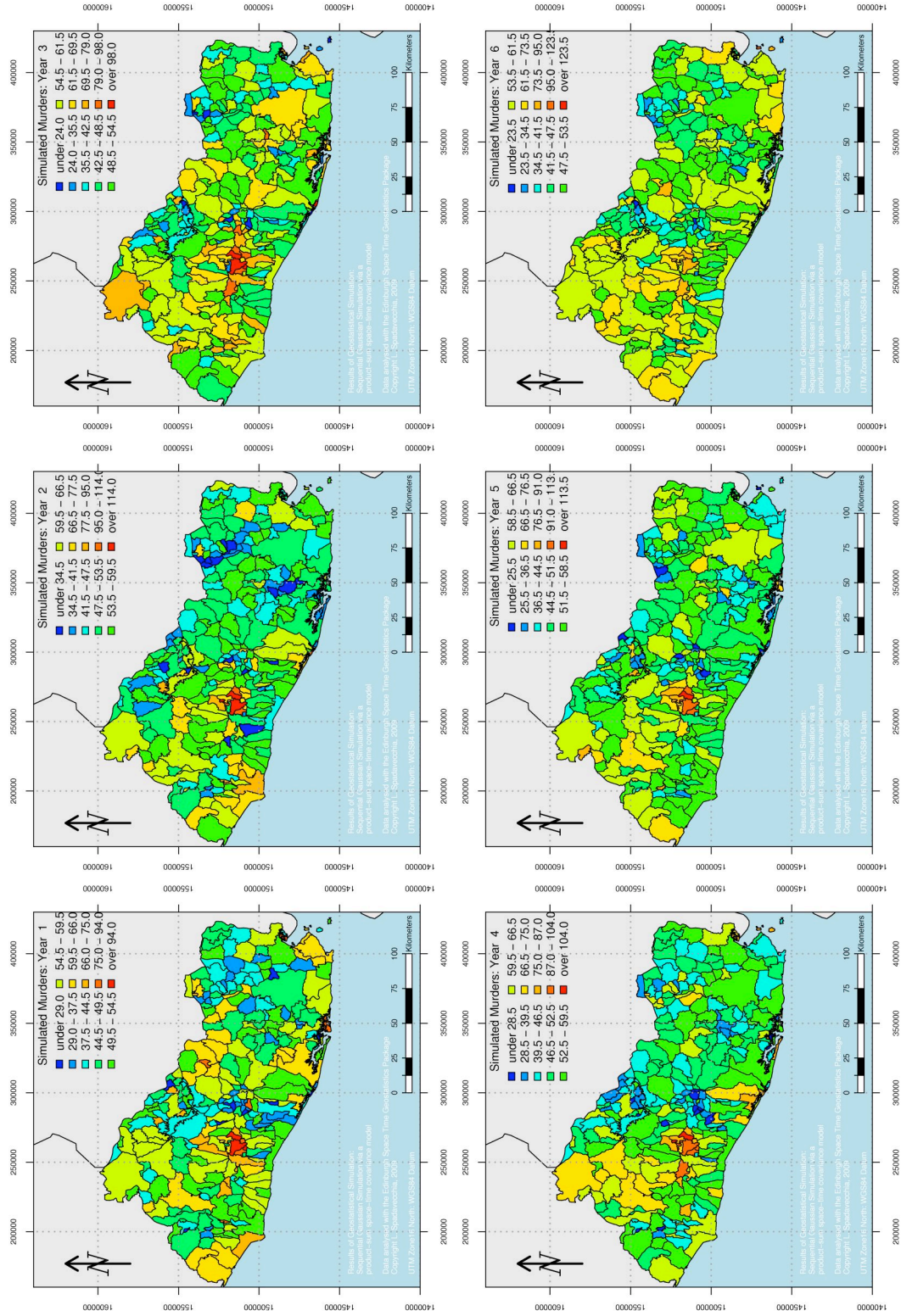


Fig 9: Simulated murder rates for El Salvadorian provinces over 6 years (Simulation 3)

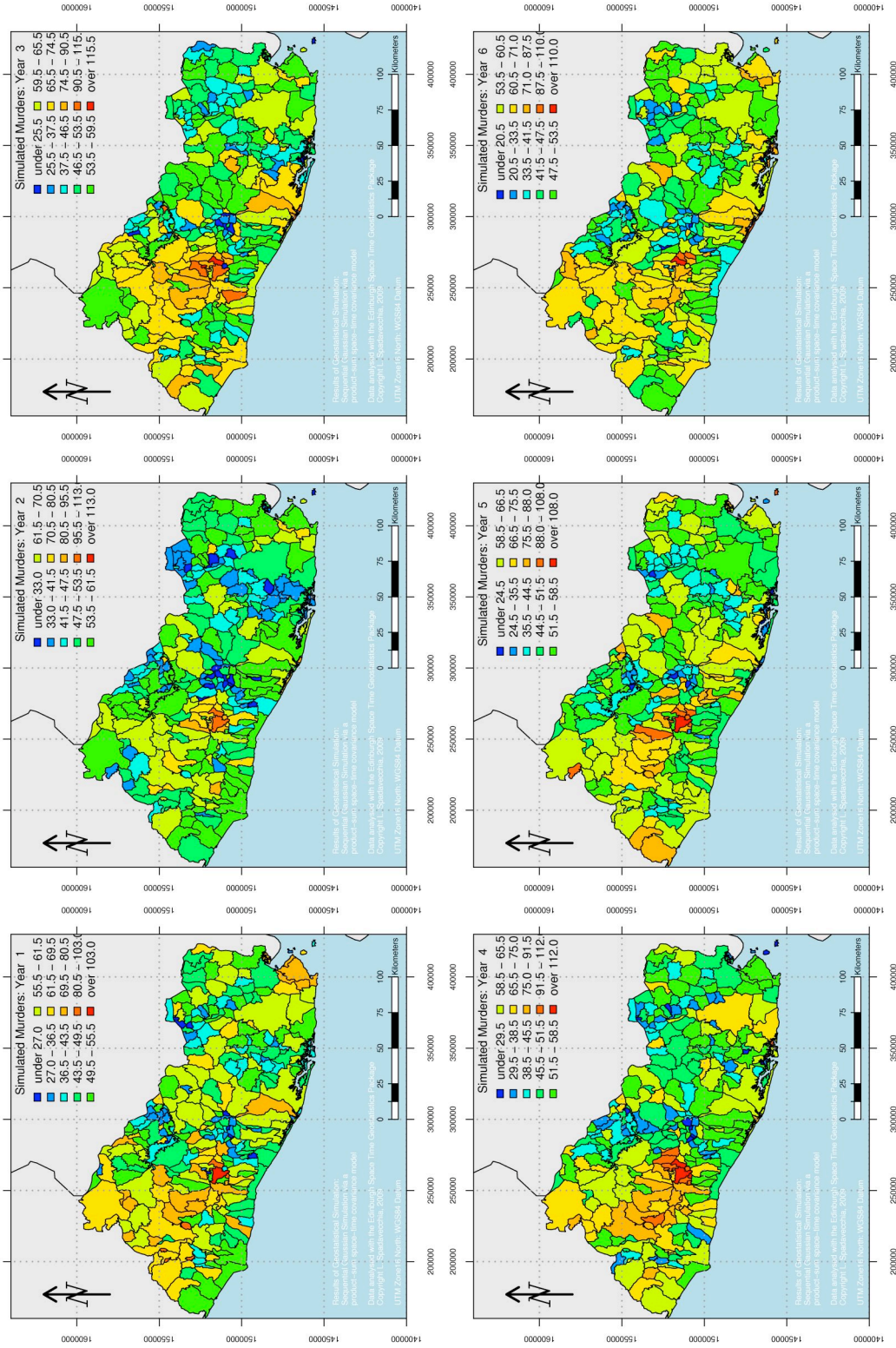


Fig 10: Simulated murder rates for El Salvadorian provinces over 6 years (Simulation 4)

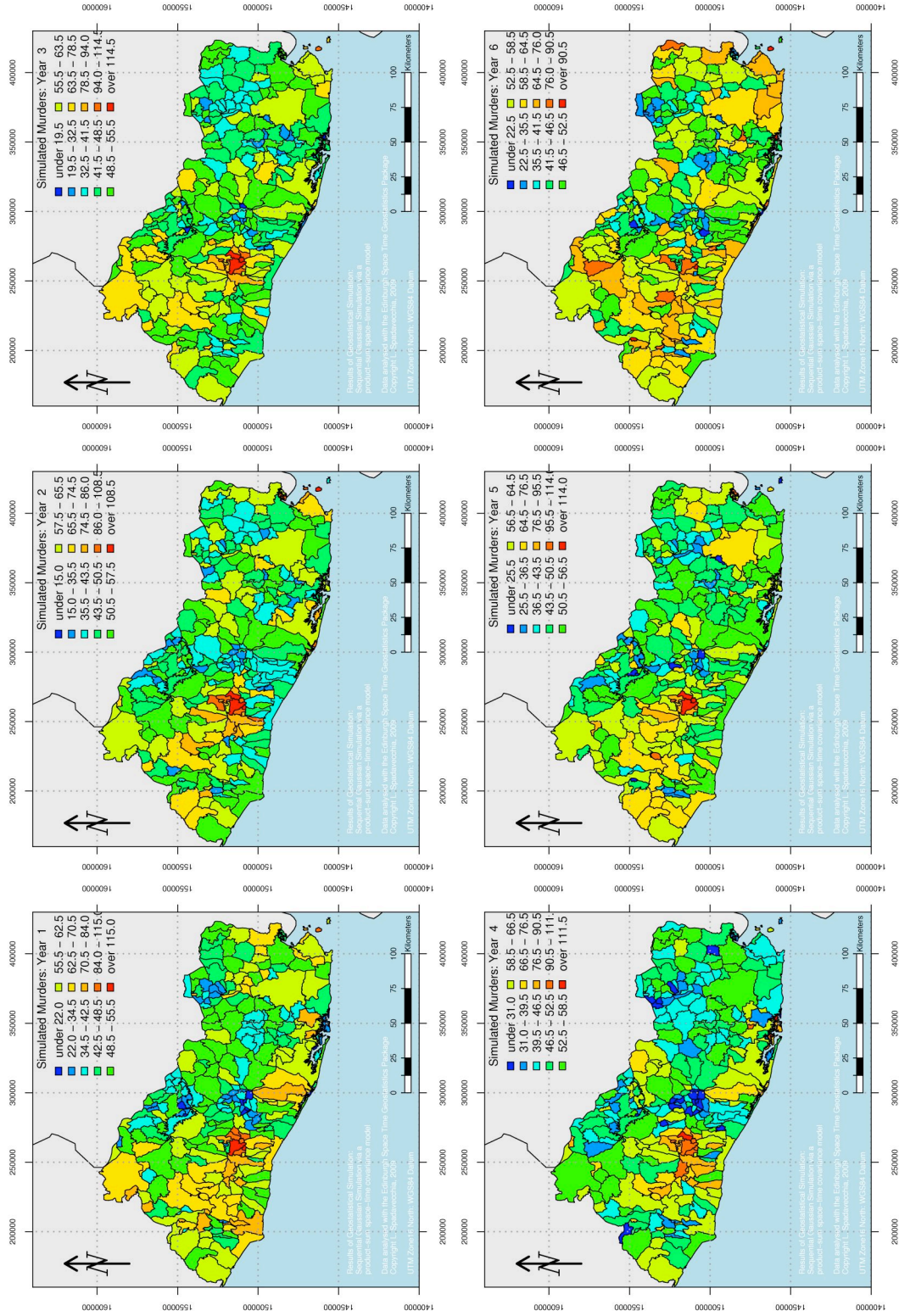


Fig 11: Simulated murder rates for El Salvadorian provinces over 6 years (Simulation 5)

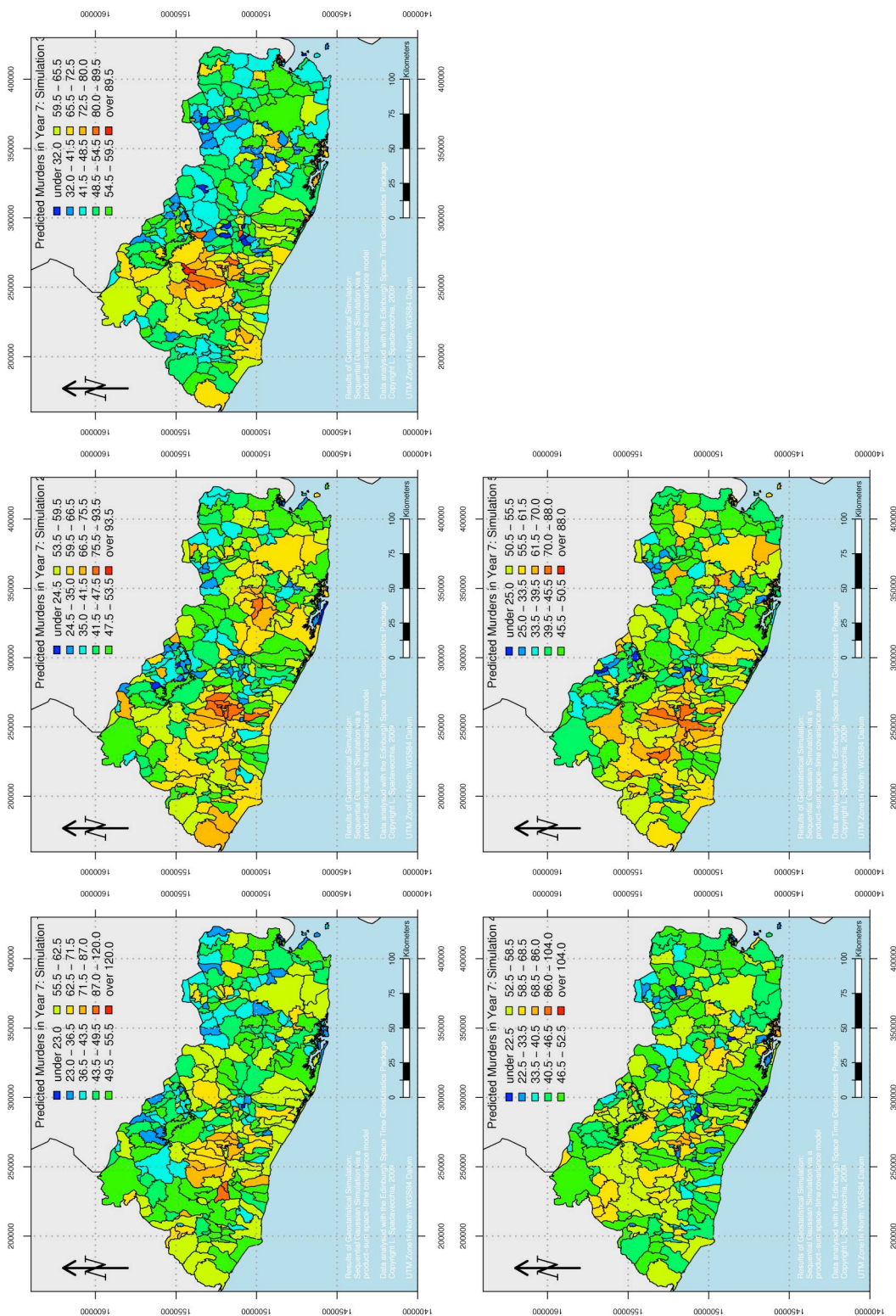


Fig 12: Five simulated realisations of murder rates for El Salvadorian provinces for year 7

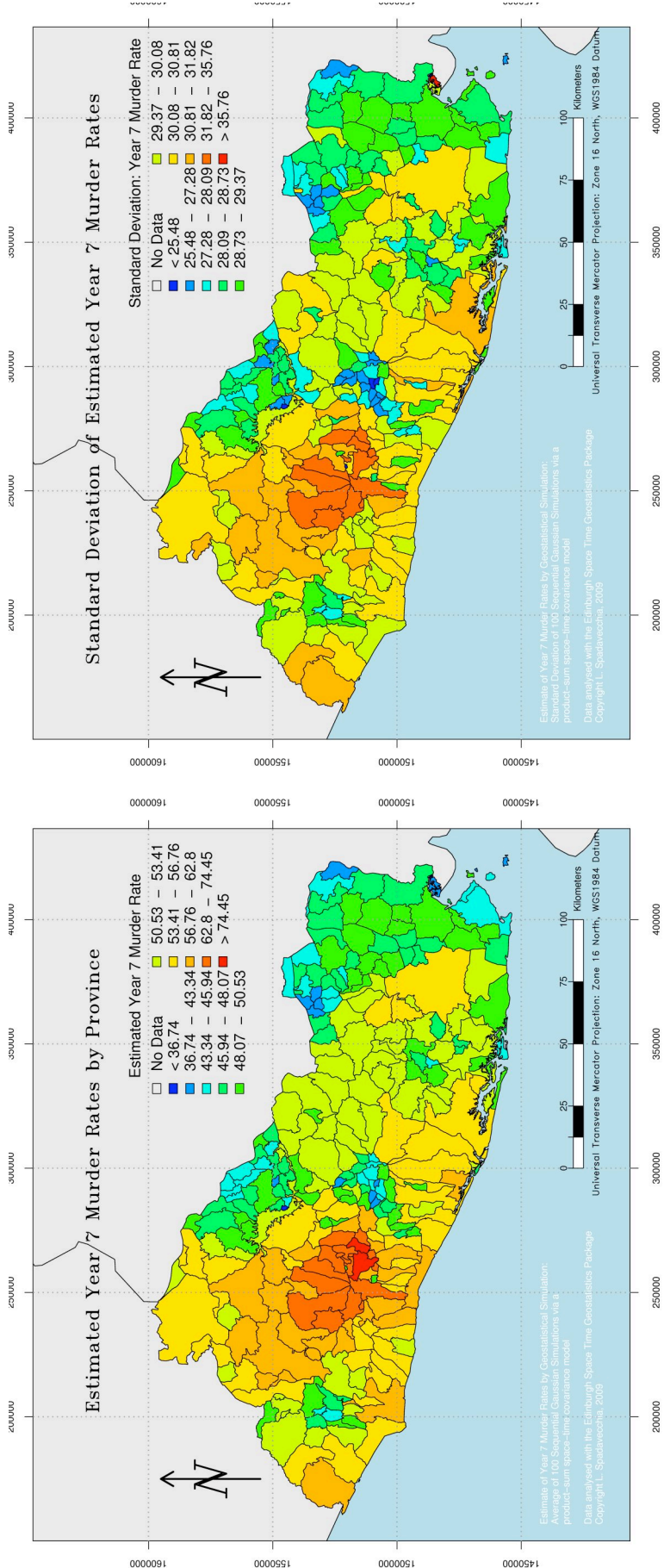


Fig 13: Estimated Murder rates and associated estimation standard deviations for El Salvadorian provinces in year 7

Notes

The data have many interesting features that present challenges for the geostatistical approach. One of the main issues is the left skew, since the underlying probability model of the method is Gaussian. Although the method is relatively insensitive to violations of distribution, the abundance of low values with rare extreme events does require some special treatment.

In order to make the simulations above I had to slightly modify the original GAUSSIM code. Since the murder rates can only be positive, I have enforced the simulations to be within the data range of the observations by disallowing negative kriging weights. The approach is the Deutsch convex estimator corrections, and is detailed in Deutsch, C.V. (1996). *Computers & Geosciences* **22**(7): 765-773. You can see the corrections in action if you run the program in debug mode. When in debug mode the rescaled weights are displayed if any negative weights arose in the original scheme (Table 2).

You may also notice in Fig 6 that the temporal semivariogram is more or less a linear unbounded increase: Such unbounded models are very difficult to implement, and so I have approximated the linear rise with an exponential function with a suitably long range. However, the unbounded nature of the rise presents problems calculating the global sill, and mis-specification of the semivariogram can result. This is not a major issue, except that occasionally, particularly when the covariance matrix has

Table 2: Gaussim debug output detailing weights correction. Note OID of -999 and SID of *** for previously simulated points

*** Weights Correction ***								
n	Covariance	Original	Corrected					
1	-246.2892	-.0026	0.0000					
2	-216.9286	0.0309	0.0308					
3	254.1175	0.2006	0.2001					
4	-8.5016	0.1027	0.1025					
5	239.9452	0.1619	0.1615					
6	213.2421	0.1364	0.1360					
7	169.3568	0.0900	0.0898					
8	157.1332	0.0817	0.0815					
9	210.6032	0.0061	0.0060					
10	187.1104	0.0070	0.0070					
11	148.6090	0.0053	0.0052					
12	137.8965	0.0054	0.0054					
13	184.8343	0.0493	0.0492					
14	164.2530	0.0442	0.0440					
15	130.4321	0.0400	0.0399					
16	121.0436	0.0411	0.0409					

w/C hat:	-246.2892	0.0026						
Observations:								

OID	SID	Value	Drift	Distance	Angle	Time	Gamma	Weight

-999	***	0.00	0.00	46738.70	62.65	0.00	1847.2892	0.0000
-999	***	0.00	0.00	50773.66	49.57	0.00	1817.9286	0.0308
-999	***	0.00	0.00	85412.83	-50.44	0.00	1346.8826	0.2001
-999	***	0.00	0.00	108427.79	-47.68	0.00	1609.5016	0.1025
1152	192	3.00	0.00	4754.85	10.76	-1.00	1361.0548	0.1615
1074	179	3.00	0.00	6061.67	36.86	-1.00	1387.7579	0.1360
1086	181	1.00	0.00	9838.26	-89.90	-1.00	1431.6433	0.0898
1170	195	4.00	0.00	10904.16	75.06	-1.00	1443.8667	0.0815
1151	192	1.00	0.00	4753.88	10.76	-2.00	1390.3967	0.0060
1073	179	3.00	0.00	6064.20	36.85	-2.00	1413.8895	0.0070
1085	181	5.00	0.00	9840.39	-89.90	-2.00	1452.3910	0.0052
1169	195	3.00	0.00	10904.72	75.08	-2.00	1463.1035	0.0054
1150	192	1.00	0.00	4753.60	10.75	-3.00	1416.1658	0.0492
1072	179	4.00	0.00	6060.60	36.86	-3.00	1436.7469	0.0440
1084	181	3.00	0.00	9839.14	-89.90	-3.00	1470.5679	0.0399
1168	195	10.00	0.00	10902.12	75.07	-3.00	1479.9563	0.0409

Mean:		2.56	0.00	24126.91			Sum:	1.0000

Estimate:								

East	North	Drift	Time	Local MU	Estimate	Variance	St.	

347984.91	1465096.25	0.00	7.00	1.76	9.30	1605.85	40	

almost singular characteristics, a negative variance may be produced. The kriging variance is given by the global sill minus the dot product of the weights and the covariance vector of the estimation location and observed data, so if the global sill is small compared to the covariance vector, a negative variance will result. In such cases we expect the variance to be small anyway, so my solution is to be pragmatic and replace negative variances with the nugget variance (i.e. the smallest possible variance for

the set when separation in space and time is zero – the underlying noise component of the data). I have coded a warning message when this occurs, which details the coordinates of the problem point. Luckily such points are relatively rare:

```
Negative variance produced!
```

```
Problem coordinate: East = 240620.7, North = 1595364.8, Time = 7.0
```

A further modification to allow simulations at time stamps with no observations was also made, so that you can forecast rates in year 7. This works by trying to satisfy the search strategy criteria by getting the right number of points from each time window, and satisfying any deficit from the next time window; i.e. if we need 12 points from each time window up to ± 3 years, but the time window ± 2 years only has 10 available points, the extra 2 will be retrieved at time window ± 3 years. Thus the total number of points the simulation is conditioned on remains the same. You will be informed of this if it is required by a warning message I have added to the code:

```
Insufficient data at plus/minus 2 time step(s) to fulfil search  
strategy:  
Attempting to fill data deficit from subsequent time steps...
```

As a final note, the simulations presented here are probably on the high side. This is as a result of the skewed distribution, and it will probably

be sensible to transform the data somehow and use the new set to derive a covariance model to simulate from and back transform the results. I resisted this approach because, for example, the lognormal approach has many complications, not least of which is the unacceptably high estimation variances that result. This may however be a necessary sacrifice.